■ README.md — Complete Content Guide

Automatic Exhaust Fan | ESP8266 + MQ-2 Sensor

What is README.md?

README.md is the first file anyone sees when they open your GitHub repository. It explains what your project does, how it works, and how to set it up. Write it in Markdown format (.md). Below is every section you must include.

Section 1 — Project Title & Badges

- # Automatic Exhaust Fan using ESP8266 & MQ-2 Sensor
- Add shields.io badges: Arduino | ESP8266 | License: MIT | Platform: IoT
- Add a one-line tagline below the title

Section 2 — Project Description

- What it does: Detects smoke and auto-activates exhaust fan
- Why it was built: Energy saving, noise reduction, no manual intervention
- Where it is used: Kitchens, offices, restaurants, labs

Section 3 — Features

- ■ Automatic smoke detection using MQ-2 sensor
- ■ Exhaust fan auto ON/OFF via relay module
- ■ LED indicators: Green=Safe, Red=Hazard, Orange=Warning
- ■ Buzzer alarm when smoke exceeds threshold
- ■ Real-time monitoring via ThingSpeak
- ■ Energy efficient — fan runs only when needed

Section 4 — Components Used

- ESP8266 NodeMCU V3 (microcontroller)
- MQ-2 Gas/Smoke Sensor
- 5V 1-Channel Relay Module
- DC Motor (exhaust fan)
- Buzzer, LEDs (Red, Green, Orange), Resistors

Section 5 — Circuit / Block Diagram

- Embed schematic_diagram.png: `![Schematic](circuit/schematic_diagram.png)`
- Embed block_diagram.png: `![Block Diagram](circuit/block_diagram.png)`

Section 6 — Working Principle

- MQ-2 reads analog smoke level continuously
- If smoke > 500 (threshold): Red LED ON, Buzzer ON, Fan ON
- If smoke = 500 (at threshold): Orange LED ON, Fan ON
- If smoke < 500: Green LED ON, Buzzer OFF, Fan OFF

Section 7 — Pin Connections

- Include a Markdown table showing all pin mappings
- ESP8266 A0 → MQ-2 Analog Out
- ESP8266 D3 → Red LED (Hazard)
- ESP8266 D4 → Relay IN (Exhaust Fan)
- ESP8266 D5 → Buzzer
- ESP8266 D6 → Green LED (Safe)

Section 8 — Installation & Setup

- Step 1: Install Arduino IDE
- Step 2: Add ESP8266 board URL in preferences
- Step 3: Clone this repository
- Step 4: Open src/automatic_exhaust_fan.ino
- Step 5: Select Board: NodeMCU 1.0, select COM port
- Step 6: Upload the sketch
- Step 7: Open Serial Monitor at 9600 baud to test

Section 9 — Source Code Explanation

- smokeSensorPin = A0 (reads analog smoke level)
- thresh = 500 (adjustable threshold value)
- D3=Hazard LED, D4=Exhaust Relay, D5=Buzzer, D6=Safe LED
- loop() reads sensor, compares with threshold, controls outputs

Section 10 — Results & Output

- Add serial_monitor_output.png screenshot
- Add thingspeak_chart.png (gas sensor live data)
- Describe observed smoke levels and fan response

Section 11 — Future Scope

- Add temperature & humidity sensors (DHT11/DHT22)
- Fan speed control using PWM
- Mobile app push notifications
- CO2 / PM2.5 air quality sensor integration
- Building Management System (BMS) integration

Section 12 — References

- [1] Journal of Advanced Research in Applied Mechanics, 2024
- [2] IEEE Access — Smart Toilet Scheduling, 2023
- [3] Embedded.com — Device Standby Power, 2023
- [4] Journal of Automotive Engineering, 2022
- [5] JMIR Research Protocols, 2023

Section 13 — License

- Add MIT License (recommended for student/open-source projects)
- Create a LICENSE file in root of repository